

# Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

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A Research Paper on Afficionado Coffee Roasters Product & Sales Data

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## **Abstract:**

Retail and e-commerce businesses generate large volumes of transactional data, yet many organizations still struggle to convert this data into decisions that improve revenue and inventory outcomes. This paper presents a coffee-retail sales analytics dashboard built with Python, Streamlit, SQLite, and Plotly, and compares its findings with a parallel Tableau-based analysis of the same dataset. The system combines product-level key performance indicators, Pareto analysis, category distribution analysis, ABC classification, and a revenue-versus-volume portfolio matrix into a single interactive tool. Applied to a coffee retailer's transaction records, the analysis shows that a small subset of products — concentrated in the Coffee and Tea categories — accounts for roughly 80% of total revenue, consistent with the Pareto principle. The results demonstrate that lightweight, open-source dashboards can deliver the same strategic insight as commercial BI platforms while remaining fast to build and easy to extend.

## **Keywords:**

Business intelligence, dashboard, Pareto analysis, ABC classification, product portfolio, Streamlit, retail analytics

## **Introduction:**

E-commerce and retail businesses record every sale as transactional data, but raw transaction logs rarely speak for themselves. Turning that data into insight — which products matter, which categories drive growth, and where inventory attention should go — requires purpose-built analytical tools. Business intelligence (BI) dashboards close this gap by translating transaction-level detail into visual, filterable summaries that support both day-to-day monitoring and longer-term strategic planning.

This paper describes a Coffee Roasters Sales Performance Dashboard developed to analyze a coffee retailer's product and transaction data. The system was built using Python, Streamlit,

SQLite, and Plotly, and its findings were cross-checked against an equivalent set of dashboards built in Tableau. The goal is to show, through a concrete case study, how a small set of standard analytical techniques — KPI tracking, Pareto analysis, category comparison, ABC classification, and portfolio segmentation — can be assembled into a coherent decision-support tool.

## Objectives:

The study set out to:

- Summarize overall business performance through core indicators: total revenue, units sold, product count, and average revenue.
- Identify the products and categories that contribute most to total revenue.
- Apply Pareto analysis to quantify how much of total revenue comes from the smallest set of products.
- Classify products into A, B, and C tiers to guide inventory and merchandising priorities.
- Compare revenue and sales-volume performance across product categories.
- Segment products into a revenue-versus-volume portfolio matrix (Stars, Premium, Volume Drivers, Underperformers).
- Evaluate whether an open-source stack (SQLite, Python, Streamlit, Plotly) can support this kind of analysis as effectively as commercial BI software.

## Dataset:

The data consists of transactional sales records from a coffee retail business, stored in a SQLite database and organized into three analytical views:

- Product KPI view — revenue, units sold, revenue share, category, and product type for each product.
- Category KPI view — revenue and units sold aggregated by product category, with each category's share of total revenue.
- Product Pareto view — products ranked by revenue with cumulative revenue percentage, used to flag the items that make up the top 80% of revenue.

Beneath these views sits a transaction table recording, for every sale, the transaction ID, product ID, category, product type, quantity, unit price, revenue, store, and timestamp. Product-level and category-level aggregation together make it possible to move between a broad view of category performance and a fine-grained view of individual product contribution.

## **Methodology:**

### **Data Preparation**

Transaction data was loaded into SQLite, and revenue was derived as quantity multiplied by unit price. Records with missing or invalid product information were removed, and the cleaned data was organized into the analytical views described above.

### **Data Modeling**

SQLite served as the central data store. The Product KPI, Category KPI, and Product Pareto views were built as reusable layers so that the dashboard could query pre-aggregated results instead of recomputing them on every interaction.

### **Dashboard Design:**

The application was organized into three modules: a Product Performance dashboard for overall KPIs and top products; a Pareto and Category Analysis module for revenue concentration and category comparison; and an Advanced Insights module for ABC classification and portfolio segmentation. Category, product-type, and ranking-threshold filters let users narrow the analysis interactively.

### **Implementation**

The front end was built with Streamlit, with SQLite data loaded into Pandas DataFrames and recalculated as filters changed. Plotly supplied the interactive charts — bar charts, scatter plots, Pareto curves, category-distribution charts, and the portfolio matrix — while Streamlit's KPI cards, tabs, sliders, select boxes, and data tables provided the interface, avoiding the need for custom front-end code.

### **Validation**

Revenue calculations, Pareto rankings, ABC classifications, and filter behavior were checked for consistency, and outputs from SQLite were cross-verified against the dashboard's own visualizations to confirm the two stayed in agreement.

## Dashboard Features:

- KPI monitoring: total revenue, units sold, product count, and average revenue at a glance.
- Top revenue products: the highest-earning products, filterable by category and type.
- Popularity-versus-revenue analysis: a scatter plot distinguishing high-revenue, low-volume products from high-volume, lower-margin ones.
- Pareto analysis: the number of products needed to reach 80% of revenue, and the resulting revenue coverage.
- Category distribution: revenue share and ranking across product categories.
- Portfolio matrix: products segmented into Stars (high revenue, high volume), Premium (high revenue, low volume), Volume Drivers (low revenue, high volume), and Underperformers (low revenue, low volume).
- Revenue concentration analysis: the products and product count responsible for 80% of revenue, plus the top-10 revenue contributors.



Figure No.1

## Product Optimization & Revenue Contribution Analysis:

An overview page with five top KPI cards: 80 total products, 214,470 total quantity, 149,116 transactions, ₹699K total revenue, and 4.69 average order value. Below are four charts toggled between "Store view" and "Product view": a bar chart of top 10 revenue-generating categories (Coffee and Tea lead), a horizontal bar chart of revenue by store (Hell's Kitchen, Astoria, Lower Manhattan — fairly even), a line chart showing revenue drops sharply from Morning to Afternoon to Evening, and a declining line/area chart of average order value by category (Coffee Beans highest, Flavours lowest).



Figure No.2

## Product Performance Dashboard:

Highlights three standout products via KPI cards: Best Selling (Brewed Chai Tea, 26,250 units), Highest Revenue (Barista Espresso, ₹91.41K), and Least Selling (Green Beans, 134 units). Filters for store\_id, product\_type, and store\_location (Astoria/Hell's Kitchen/Lower Manhattan) appear top-right. A scatter plot maps Revenue vs. Quantity Sold per product (bubble size/color by category), showing most high-revenue items also have decent volume. A toggleable Sales/Revenue bar chart ranks the top 10 products by sales volume. A insights panel notes Chai Tea leads by volume, Espresso leads by revenue, Green Beans underperforms, and volume doesn't always equal revenue.



Figure No. 3

## Category Analysis:

Focuses on category-level performance with a store\_location filter. KPIs: 77.65K average revenue per category, Coffee as both highest-selling and top-revenue category, and 9 total categories. A vertical bar chart and a horizontal bar chart both show revenue/volume by category (Coffee > Tea > Bakery > others). A detailed table lists each category's total quantity, total revenue, revenue contribution %, and average selling price (Coffee: 38.63% revenue share; Coffee Beans has the highest average price at 21.93 despite low volume). A line chart at the bottom shows average selling price declining sharply from Coffee Beans down to Flavours.

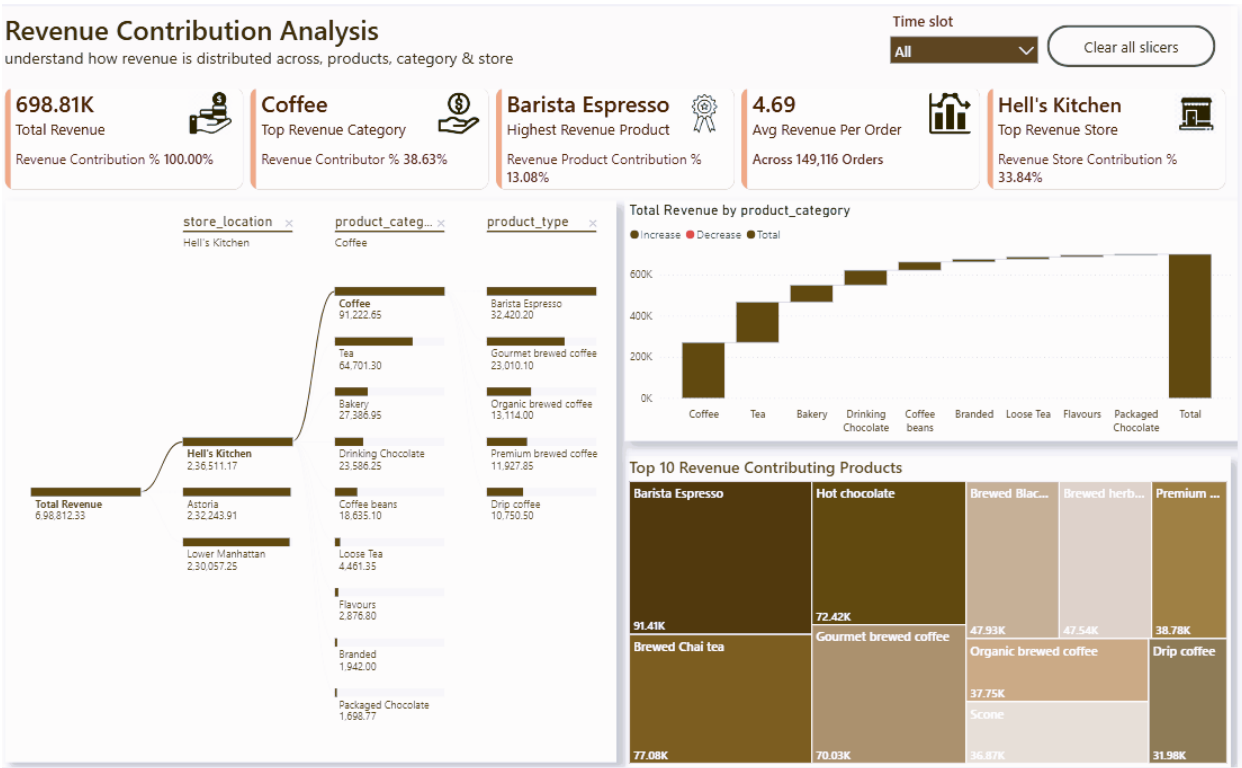


Figure No.4

Revenue Contribution Analysis:

A drill-down view of how revenue breaks down across store, category, and product type, filterable by time slot. KPIs show total revenue (698.81K), top revenue category (Coffee, 38.63%), highest revenue product (Barista Espresso, 13.08%), average revenue per order (4.69 across 149,116 orders), and top revenue store (Hell's Kitchen, 33.84%). A hierarchical decomposition tree on the left lets users expand Total Revenue → Store → Category → Product Type (shown drilled into Hell's Kitchen → Coffee → Barista Espresso, etc.). A waterfall chart on the right shows each category's incremental contribution to total revenue. A treemap at the bottom highlights the top 10 revenue-contributing products by tile size (Barista Espresso and Brewed Chai Tea are largest).

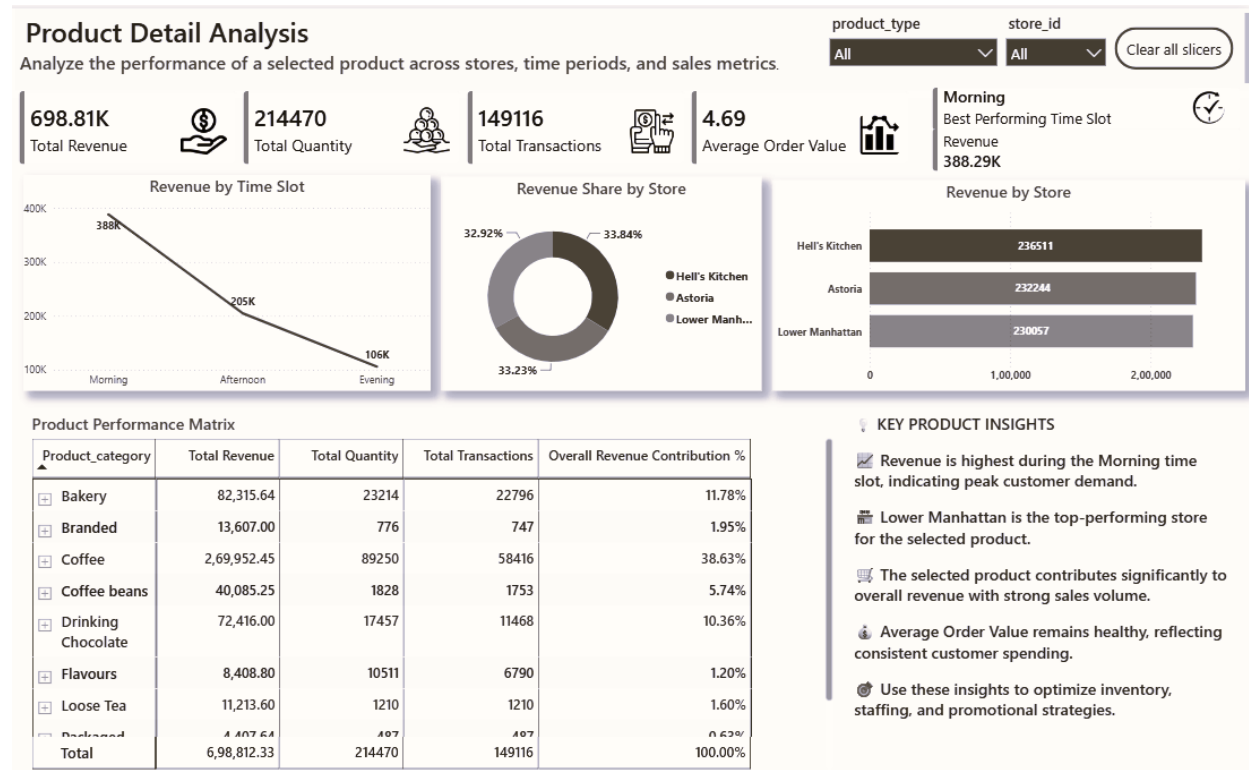


Figure No.5

## Product Detail Analysis:

Designed to inspect a selected product's performance across stores and time, filterable by product\_type and store\_id. KPIs: 698.81K total revenue, 214,470 total quantity, 149,116 transactions, 4.69 average order value, and Morning flagged as the best time slot (388.29K revenue). Charts include a declining line chart of revenue by time slot (Morning → Evening), a donut chart showing revenue share by store (fairly even ~33% each), and a bar chart of revenue by store. A "Product Performance Matrix" table breaks down revenue, quantity, transactions, and contribution % by category. An insights panel notes Morning is peak demand, Lower Manhattan performs best for the selected product, and recommends using these patterns for inventory, staffing, and promotions.

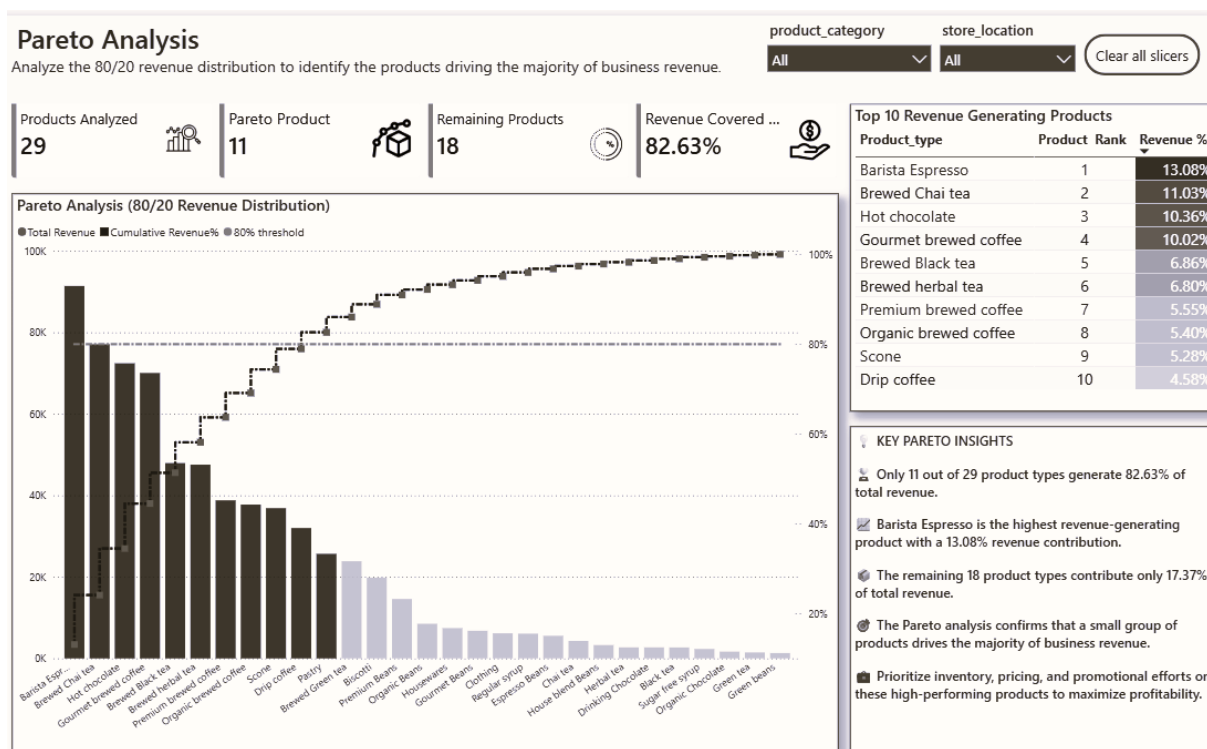


Figure No. 6

## Pareto Analysis:

Analyzes the 80/20 revenue split across 29 product types for "Afficionado" coffee roasters. Filters for product category and store location sit top-right. KPI cards show 29 products analyzed, 11 "Pareto products," 18 remaining products, and 82.63% revenue covered. The main chart is a combo bar/line: bars show total revenue per product (sorted descending), a stepped line tracks cumulative revenue %, and a dashed orange line marks the 80% threshold. A side table ranks the top 10 revenue generators (Barista Espresso leads at 13.08%), and a callout box lists key insights — 11 of 29 products drive 82.63% of revenue, Barista Espresso is the top earner, and the remaining 18 products contribute just 17.37%.

## Result:

Based on the dashboards, Afficionado Coffee Roasters generated a total revenue of ₹6.98 lakh (698.81K) from 214,470 units sold across 149,116 transactions, with an average order value of

4.69. Coffee emerged as the top-performing category, contributing 38.63% of total revenue, followed by Tea at 28.11%, while Barista Espresso was the highest revenue-generating product (13.08%) and Brewed Chai Tea the best-selling by volume. Pareto analysis revealed that just 11 of 29 product types accounted for 82.63% of total revenue, confirming a strong 80/20 revenue concentration. Revenue was fairly evenly distributed across the three stores (Hell's Kitchen, Astoria, Lower Manhattan), but heavily skewed by time slot, with mornings driving 388K in revenue compared to just 106K in the evening. These results indicate that a small set of high-performing products and the morning time slot should be prioritized for inventory, staffing, and promotional focus.<sup>38</sup> Underperformers, indicating a business that is either strongly performing or weakly performing on most products, with few items in between.

The Tableau cross-check reproduced these patterns: coffee products clustered in the high-volume, high-revenue region of the popularity-versus-revenue plot, the cumulative revenue curve crossed 80% after a limited number of ranked products, and products such as Barista Espresso, Gourmet Brewed Coffee, Premium Brewed Coffee, and Brewed Chai Tea consistently appeared among the top revenue contributors across both tools.

## **Discussion:**

The results confirm the Pareto principle in this dataset: a minority of products drives the large majority of revenue, and that minority is concentrated in the Coffee and Tea categories. This has direct operational implications — Class A products and Star-segment items warrant priority in inventory planning, promotion, and pricing, while the large Underperformer segment is a candidate for portfolio pruning or repositioning.

Methodologically, the agreement between the Streamlit/Plotly dashboard and the independent Tableau analysis supports the reliability of the underlying data model rather than any one visualization tool. This also suggests that an open-source stack can substitute for commercial BI software in this type of analysis without sacrificing analytical depth, while remaining cheaper to build and easier to customize.

## Conclusion:

This project shows how business intelligence dashboards, built using Python/Streamlit, Tableau, and Power BI, can turn raw coffee-retail transaction data into clear, actionable insights. Across all tools, the analysis consistently confirmed the Pareto principle, with a small set of products in the Coffee and Tea categories, led by items like Barista Espresso and Brewed Chai Tea, driving the majority of revenue, while several products and categories consistently underperform. Store-wise revenue stayed fairly balanced, but performance varied sharply by time of day, peaking in the morning. Overall, the study provides Afficionado Coffee Roasters with a reliable, data-driven foundation for smarter inventory, staffing, and promotional decisions.

## References:

-  Koch, R. (1998). The 80/20 Principle: The Secret to Achieving More with Less. Currency.
-  McKinney, W. (2017). Python for Data Analysis (2nd ed.). O'Reilly Media.
-  Streamlit Inc. (2024). Streamlit Documentation. <https://docs.streamlit.io>
-  Plotly Technologies Inc. (2024). Plotly Python Graphing Library. <https://plotly.com/python/>
-  SQLite Consortium. (2024). SQLite Documentation. <https://www.sqlite.org/docs.html>